

2006 年数学(三)真题解析

一、填空题

(1)【答案】 1.

【解】 当 n 为偶数时, $\lim_{n \rightarrow \infty} \left(\frac{n+1}{n}\right)^{(-1)^n} = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right) = 1$;

当 n 为奇数时, $\lim_{n \rightarrow \infty} \left(\frac{n+1}{n}\right)^{(-1)^n} = \lim_{n \rightarrow \infty} \left(\frac{n+1}{n}\right)^{-1} = \lim_{n \rightarrow \infty} \frac{n}{n+1} = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n+1}\right) = 1$;

所以 $\lim_{n \rightarrow \infty} \left(\frac{n+1}{n}\right)^{(-1)^n} = 1$.

(2)【答案】 $2e^3$.

【解】 由 $f(2) = 1$, 得 $f'(2) = e$.

又由 $f'(x) = e^{f(x)}$, 得 $f''(x) = e^{f(x)} f'(x) = e^{2f(x)}$, $f'''(x) = 2e^{2f(x)} f'(x) = 2e^{3f(x)}$,
于是 $f'''(2) = 2e^3$.

(3)【答案】 $4dx - 2dy$.

【解】 $\frac{\partial z}{\partial x} = 8xf'(4x^2 - y^2)$, $\frac{\partial z}{\partial y} = -2yf'(4x^2 - y^2)$,

$$\left. \frac{\partial z}{\partial x} \right|_{(1,2)} = 8f'(0) = 4, \quad \left. \frac{\partial z}{\partial y} \right|_{(1,2)} = -4f'(0) = -2,$$

于是 $dz|_{(1,2)} = 4dx - 2dy$.

(4)【答案】 2.

【解】 由 $BA = B + 2E$, 得 $B(A - E) = 2E$, 两边取行列式, 得 $|B| \cdot |A - E| = 4$.

因为 $A - E = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$, 所以 $|A - E| = 2$, 于是 $|B| = 2$.

(5)【答案】 $\frac{1}{9}$.

【解】 由 $X \sim U(0, 3)$, $Y \sim U(0, 3)$ 得 X, Y 的边缘密度为

$$f_X(x) = \begin{cases} \frac{1}{3}, & 0 < x < 3, \\ 0, & \text{其他.} \end{cases} \quad f_Y(y) = \begin{cases} \frac{1}{3}, & 0 < y < 3, \\ 0, & \text{其他.} \end{cases}$$

由 X, Y 独立得

$$\begin{aligned} P\{\max(X, Y) \leq 1\} &= P\{X \leq 1, Y \leq 1\} = P\{X \leq 1\}P\{Y \leq 1\} \\ &= \int_0^1 \frac{1}{3} dx \cdot \int_0^1 \frac{1}{3} dy = \frac{1}{9}. \end{aligned}$$

(6)【答案】 2.

【解】 $E(X) = \int_{-\infty}^{+\infty} xf(x)dx = 0$,

$$E(X^2) = \int_{-\infty}^{+\infty} x^2 f(x)dx = \int_0^{+\infty} x^2 e^{-x} dx = \Gamma(3) = 2.$$

于是 $E(S^2) = D(X) = 2$.

方法点评: 本题考查样本方差的数字特征. 本题需要熟练掌握如下性质:

(1) 设 $X_1, X_2, \dots, X_n$ 为来自总体 X 的简单随机样本, $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$, 则

$$E(S^2) = D(X).$$

(2) Γ 函数的定义为 $\Gamma(\alpha) = \int_0^{+\infty} x^{\alpha-1} e^{-x} dx$, 且具有性质:

$$\Gamma(\alpha+1) = \alpha \Gamma(\alpha); \quad \Gamma(n+1) = n!; \quad \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}.$$

二、选择题

(7) 【答案】 (A).

【解】 方法一 $dy = f'(x)\Delta x$, $\Delta y = f(x+\Delta x) - f(x) = f'(\xi)\Delta x$ ($x < \xi < x+\Delta x$).

因为 $f''(x) > 0$, 所以 $f'(x)$ 单调增加, 于是 $0 < f'(x) < f'(\xi)$.

再由 $\Delta x > 0$, 得 $0 < f'(x)\Delta x < f'(\xi)\Delta x$, 即 $0 < dy < \Delta y$, 应选(A).

方法二 由泰勒公式得

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(\xi)}{2!}(x-x_0)^2, \text{ 其中 } \xi \text{ 介于 } x_0 \text{ 与 } x \text{ 之间,}$$

因为 $f''(x) > 0$, 所以 $f(x) - f(x_0) \geq f'(x_0)(x-x_0)$, 当且仅当 $x = x_0$ 时等号成立, 故 $\Delta y \geq dy$.

因为 $f'(x_0) > 0$, $\Delta x = x - x_0 > 0$, 所以 $dy = f'(x_0)(x-x_0) > 0$,

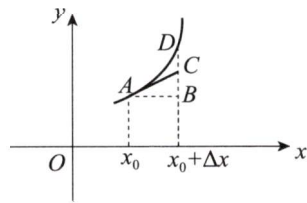
于是 $\Delta y > dy > 0$, 应选(A).

方法三 $\Delta y = f(x_0 + \Delta x) - f(x_0) = f'(\xi)\Delta x$ ($x_0 < \xi < x_0 + \Delta x$),

则 $\Delta y - dy = [f'(\xi) - f'(x_0)]\Delta x = f''(\eta)(\xi - x_0)\Delta x$ ($x_0 < \eta < \xi$),

由 $f''(x) > 0$ 得 $\Delta y - dy > 0$, 即 $\Delta y > dy$, 又 $dy > 0$, 故 $\Delta y > dy > 0$, 应选(A).

方法四 因为 $f'(x) > 0$, $f''(x) > 0$, 所以 $y = f(x)$ 为单调增加的凹函数, 如图所示. 因为 $\Delta x > 0$, 所以 $dy = |BC| > 0$, $\Delta y = |BD| > |BC|$, 应选(A).



二(7) 题图

(8) 【答案】 (C).

【解】 由 $\lim_{h \rightarrow 0} \frac{f(h^2)}{h^2} = 1$, 得 $\lim_{h \rightarrow 0} f(h^2) = 0$, 因为 $f(x)$ 在 $x=0$ 处

连续, 所以 $f(0) = 0$.

又由 $\lim_{h \rightarrow 0} \frac{f(h^2)}{h^2} = \lim_{h \rightarrow 0} \frac{f(h^2) - f(0)}{h^2} = f'_+(0)$, 得 $f'_+(0) = 1$, 应选(C).

(9) 【答案】 (D).

【解】 方法一 令 $S_n = a_1 + a_2 + \dots + a_n$, 因为 $\sum_{n=1}^{\infty} a_n$ 收敛, 所以 $\lim_{n \rightarrow \infty} a_n = 0$ 且 $\lim_{n \rightarrow \infty} S_n$ 存在,

令 $\lim_{n \rightarrow \infty} S_n = S$,

令 $S_n^{(1)} = \frac{a_1 + a_2}{2} + \frac{a_2 + a_3}{2} + \dots + \frac{a_n + a_{n+1}}{2} = S_n - \frac{a_1}{2} + \frac{a_{n+1}}{2}$,

因为 $\lim_{n \rightarrow \infty} S_n^{(1)} = S - \frac{a_1}{2}$ 存在, 所以 $\sum_{n=1}^{\infty} \frac{a_n + a_{n+1}}{2}$ 收敛, 应选(D).

方法二 取 $a_n = \frac{(-1)^n}{n}$, 级数 $\sum_{n=1}^{\infty} a_n$ 收敛, 但 $\sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{\infty} \frac{1}{n}$ 及 $\sum_{n=1}^{\infty} (-1)^n a_n = \sum_{n=1}^{\infty} \frac{1}{n}$ 发散, (A) 不正确;

取 $a_n = \frac{(-1)^n}{\sqrt{n}}$, 级数 $\sum_{n=1}^{\infty} a_n$ 收敛, $\sum_{n=1}^{\infty} a_n a_{n+1} = -\sum_{n=1}^{\infty} \frac{1}{\sqrt{n(n+1)}}$,

因为 $\frac{1}{\sqrt{n(n+1)}} \sim \frac{1}{n}$ 且级数 $\sum_{n=1}^{\infty} \frac{1}{n}$ 发散, 所以 $-\sum_{n=1}^{\infty} \frac{1}{\sqrt{n(n+1)}}$ 发散, 即 $\sum_{n=1}^{\infty} a_n a_{n+1}$ 发散,

(C) 不正确, 应选(D).

方法三 由级数 $\sum_{n=1}^{\infty} a_n$ 收敛得 $\sum_{n=1}^{\infty} \frac{1}{2} a_n$ 与 $\sum_{n=1}^{\infty} \frac{1}{2} a_{n+1}$ 收敛,

由级数收敛的基本性质得 $\sum_{n=1}^{\infty} \frac{a_n + a_{n+1}}{2}$ 收敛.

(10)【答案】(B).

【解】 $y_1(x) - y_2(x)$ 为齐次线性微分方程 $y' + P(x)y = 0$ 的解,

$y' + P(x)y = Q(x)$ 的通解为 $y' + P(x)y = 0$ 的通解与 $y' + P(x)y = Q(x)$ 的特解之和,

故 $y' + P(x)y = Q(x)$ 的通解为 $y_1(x) + C[y_1(x) - y_2(x)]$, 应选(B).

(11)【答案】(D).

【解】 方法一 因为 $\varphi'_y(x, y) \neq 0$, 所以 $\varphi(x, y) = 0$ 确定 y 为 x 的函数, 设为 $y = y(x)$, 代入 $z = f(x, y)$ 中, 得 $z = f[x, y(x)]$.

因为 (x_0, y_0) 为极值点, 所以 $\left. \frac{dz}{dx} \right|_{x=x_0} = f'_x(x_0, y_0) + f'_y(x_0, y_0) \cdot y'(x_0) = 0$,

若 $f'_x(x_0, y_0) \neq 0$, 则 $f'_y(x_0, y_0) \neq 0$, 应选(D).

方法二 令 $F(x, y, \lambda) = f(x, y) + \lambda \varphi(x, y)$, 因为 (x_0, y_0) 为极值点, 所以

$$\begin{cases} F'_x(x_0, y_0) = f'_x(x_0, y_0) + \lambda \varphi'_x(x_0, y_0) = 0, \\ F'_y(x_0, y_0) = f'_y(x_0, y_0) + \lambda \varphi'_y(x_0, y_0) = 0, \\ F'_\lambda(x_0, y_0) = \varphi(x_0, y_0) = 0. \end{cases}$$

当 $f'_x(x_0, y_0) \neq 0$ 时, $\lambda \neq 0$.

由 $\varphi'_y(x_0, y_0) \neq 0$, 得 $\lambda \varphi'_y(x_0, y_0) \neq 0$, 则 $f'_y(x_0, y_0) \neq 0$, 应选(D).

(12)【答案】(A).

【解】 方法一 令 $Q = (\alpha_1, \alpha_2, \dots, \alpha_s)$, $(A\alpha_1, A\alpha_2, \dots, A\alpha_s) = AQ$,

则 $r(A\alpha_1, A\alpha_2, \dots, A\alpha_s) = r(AQ) \leq r(Q)$.

若 $\alpha_1, \alpha_2, \dots, \alpha_s$ 线性相关, 则 $r(Q) = r(\alpha_1, \alpha_2, \dots, \alpha_s) < s$,

于是 $r(A\alpha_1, A\alpha_2, \dots, A\alpha_s) \leq r(Q) < s$,

即 $A\alpha_1, A\alpha_2, \dots, A\alpha_s$ 线性相关, 应选(A).

方法二 若 $\alpha_1, \alpha_2, \dots, \alpha_s$ 线性相关, 则存在不全为零的常数 $k_1, k_2, \dots, k_s$, 使得

$$k_1 \alpha_1 + k_2 \alpha_2 + \dots + k_s \alpha_s = 0,$$

等式两边左乘 A 得

$$k_1 A\alpha_1 + k_2 A\alpha_2 + \dots + k_s A\alpha_s = 0,$$

由线性相关的定义得 $A\alpha_1, A\alpha_2, \dots, A\alpha_s$ 线性相关, 应选(A).

(13)【答案】 (B).

$$\text{【解】 } B = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} A = PA, C = B \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = BP^{-1},$$

于是 $C = PAP^{-1}$, 应选(B).

(14)【答案】 (A).

【解】 由 $X \sim N(\mu_1, \sigma_1^2)$, $Y \sim N(\mu_2, \sigma_2^2)$, 得 $\frac{X-\mu_1}{\sigma_1} \sim N(0,1)$, $\frac{Y-\mu_2}{\sigma_2} \sim N(0,1)$,

$$P\{|X-\mu_1| < 1\} = P\left\{-\frac{1}{\sigma_1} < \frac{X-\mu_1}{\sigma_1} < \frac{1}{\sigma_1}\right\} = \Phi\left(\frac{1}{\sigma_1}\right) - \Phi\left(-\frac{1}{\sigma_1}\right) = 2\Phi\left(\frac{1}{\sigma_1}\right) - 1,$$

$$P\{|Y-\mu_2| < 1\} = P\left\{-\frac{1}{\sigma_2} < \frac{Y-\mu_2}{\sigma_2} < \frac{1}{\sigma_2}\right\} = \Phi\left(\frac{1}{\sigma_2}\right) - \Phi\left(-\frac{1}{\sigma_2}\right) = 2\Phi\left(\frac{1}{\sigma_2}\right) - 1,$$

由 $P\{|X-\mu_1| < 1\} > P\{|Y-\mu_2| < 1\}$, 得 $\Phi\left(\frac{1}{\sigma_1}\right) > \Phi\left(\frac{1}{\sigma_2}\right)$, 即 $\frac{1}{\sigma_1} > \frac{1}{\sigma_2}$ 或 $\sigma_1 < \sigma_2$,

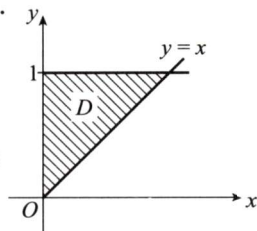
应选(A).

三、解答题

$$\begin{aligned} (15) \text{【解】 } (I) g(x) &= \lim_{y \rightarrow +\infty} f(x, y) = \lim_{y \rightarrow +\infty} \left(\frac{y}{1+xy} - \frac{1-y \sin \frac{\pi x}{y}}{\arctan x} \right) \\ &= \frac{1}{x} - \frac{1}{\arctan x} \lim_{y \rightarrow +\infty} \left(1 - \pi x \frac{\sin \frac{\pi x}{y}}{\frac{\pi x}{y}} \right) = \frac{1}{x} - \frac{1-\pi x}{\arctan x}. \end{aligned}$$

$$\begin{aligned} (II) \lim_{x \rightarrow 0^+} g(x) &= \lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1-\pi x}{\arctan x} \right) = \lim_{x \rightarrow 0^+} \frac{\arctan x - x(1-\pi x)}{x \arctan x} \\ &= \lim_{x \rightarrow 0^+} \frac{\arctan x - x + \pi x^2}{x^2} = \lim_{x \rightarrow 0^+} \frac{\frac{1}{1+x^2} - 1 + 2\pi x}{2x} \\ &= \lim_{x \rightarrow 0^+} \frac{-\frac{x^2}{1+x^2} + 2\pi x}{2x} = \frac{1}{2} \lim_{x \rightarrow 0^+} \left(2\pi - \frac{x}{1+x^2} \right) = \pi. \end{aligned}$$

$$\begin{aligned} (16) \text{【解】 } \iint_D \sqrt{y^2 - xy} \, dx \, dy &= \int_0^1 dy \int_0^y \sqrt{y^2 - xy} \, dx \\ &= -\int_0^1 \frac{1}{y} dy \int_0^y \sqrt{y^2 - xy} \, d(y^2 - xy) \\ &= -\int_0^1 \frac{1}{y} \cdot \frac{2}{3} (y^2 - xy)^{\frac{3}{2}} \Big|_0^y dy \\ &= \frac{2}{3} \int_0^1 y^2 dy = \frac{2}{9}. \end{aligned}$$



三(16)题图

(17)【证明】 令 $f(x) = x \sin x + 2 \cos x + \pi x - a \sin a - 2 \cos a - \pi a$, $f(a) = 0$.

$$f'(x) = x \cos x - \sin x + \pi,$$

因为 $f''(x) = \cos x - x \sin x - \cos x = -x \sin x < 0$ ($0 < x < \pi$),

所以 $f'(x)$ 单调减少, 又因为 $f'(\pi) = 0$, 所以 $f'(x) > 0 (0 < x < \pi)$,
故 $f(x)$ 在 $[0, \pi]$ 上单调增加, 因为 $b > a$ 且 $f(a) = 0$, 所以 $f(b) > 0$, 即

$$b \sin b + 2 \cos b + \pi b > a \sin a + 2 \cos a + \pi a.$$

方法点评: 本题考查利用单调性证明不等式.

证明关于两个常数的不等式时, 一般先将其中一个常数改为 x 构造辅助函数, 然后利用单调性证明.

(18)【解】 (I) 设曲线方程为 $L: y = y(x)$, 直线 OP 的斜率为 $\frac{y}{x}$, 由题意得

$$\frac{dy}{dx} - \frac{y}{x} = ax \quad \text{或} \quad \frac{dy}{dx} - \frac{1}{x}y = ax,$$

解得 $y = \left(a \int x e^{\int -\frac{1}{x} dx} dx + C \right) e^{-\int \frac{1}{x} dx} = (ax + C)x = Cx + ax^2,$

由 $y(1) = 0$, 得 $C = -a$, 故 $y = y(x) = ax^2 - ax$.

(II) 由 $\begin{cases} y = ax, \\ y = ax^2 - ax, \end{cases}$ 得 $x = 0, x = 2$,

由 $\int_0^2 [ax - (ax^2 - ax)] dx = a \int_0^2 (2x - x^2) dx = \frac{4}{3}a = \frac{8}{3}$, 得 $a = 2$.

(19)【解】 方法一 由 $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = 1$, 得 $R = 1$,

当 $x = \pm 1$ 时, 因为 $\sum_{n=1}^{\infty} \left| \frac{(-1)^{n-1} (\pm 1)^{2n+1}}{n(2n-1)} \right| = \sum_{n=1}^{\infty} \frac{1}{n(2n-1)}$ 收敛, 即当 $x = \pm 1$ 时, 级数绝对收敛, 故级数的收敛域为 $[-1, 1]$.

令 $S(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^{2n+1}}{n(2n-1)},$

$S(x) = x \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^{2n}}{n(2n-1)} = x S_1(x)$, 其中 $S_1(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^{2n}}{n(2n-1)}, S_1(0) = 0,$

$S'_1(x) = 2 \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^{2n-1}}{2n-1}, S'_1(0) = 0,$

$S''_1(x) = 2 \sum_{n=1}^{\infty} (-1)^{n-1} x^{2n-2} = 2 \sum_{n=1}^{\infty} (-x^2)^{n-1} = \frac{2}{1+x^2},$

$S'_1(x) = S'_1(x) - S'_1(0) = \int_0^x \frac{2}{1+x^2} dx = 2 \arctan x,$

于是 $S_1(x) = S_1(x) - S_1(0) = \int_0^x 2 \arctan x dx$

$= 2x \arctan x \Big|_0^x - \int_0^x \frac{2x}{1+x^2} dx = 2x \arctan x - \ln(1+x^2),$

故 $S(x) = 2x^2 \arctan x - x \ln(1+x^2) (-1 \leq x \leq 1).$

方法二 由 $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = 1$ 得 $R = 1$,

当 $x = \pm 1$ 时, $\sum_{n=1}^{\infty} \left| \frac{(-1)^{n-1} (\pm 1)^{2n+1}}{n(2n-1)} \right| = \sum_{n=1}^{\infty} \frac{1}{n(2n-1)}$ 收敛, 故收敛域为 $[-1, 1]$.

$$S(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^{2n+1}}{n(2n-1)} = 2 \left[\sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^{2n+1}}{2n-1} - \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^{2n+1}}{2n} \right],$$

$$\begin{aligned} \text{而 } \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^{2n+1}}{2n-1} &= x^2 \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^{2n-1}}{2n-1} = x^2 \int_0^x \left[\sum_{n=1}^{\infty} (-1)^{n-1} x^{2n-2} \right] dx \\ &= x^2 \int_0^x \frac{1}{1+x^2} dx = x^2 \arctan x; \end{aligned}$$

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^{2n+1}}{2n} &= x \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^{2n}}{2n} = x \int_0^x \left[\sum_{n=1}^{\infty} (-1)^{n-1} x^{2n-1} \right] dx \\ &= x \int_0^x \frac{x}{1+x^2} dx = \frac{x}{2} \ln(1+x^2), \end{aligned}$$

故 $S(x) = 2x^2 \arctan x - x \ln(1+x^2) (-1 \leq x \leq 1)$.

(20)【解】 向量组 $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ 线性相关的充分必要条件是 $|\alpha_1, \alpha_2, \alpha_3, \alpha_4| = 0$,

$$\text{而 } |\alpha_1, \alpha_2, \alpha_3, \alpha_4| = \begin{vmatrix} 1+a & 2 & 3 & 4 \\ 1 & 2+a & 3 & 4 \\ 1 & 2 & 3+a & 4 \\ 1 & 2 & 3 & 4+a \end{vmatrix} = a^3(a+10),$$

当 $a = -10$ 或 $a = 0$ 时, 向量组 $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ 线性相关.

当 $a = 0$ 时, 极大线性无关组为 α_1 , 且 $\alpha_2 = 2\alpha_1, \alpha_3 = 3\alpha_1, \alpha_4 = 4\alpha_1$;

当 $a = -10$ 时, 由

$$\begin{aligned} (\alpha_1, \alpha_2, \alpha_3, \alpha_4) &= \begin{pmatrix} -9 & 2 & 3 & 4 \\ 1 & -8 & 3 & 4 \\ 1 & 2 & -7 & 4 \\ 1 & 2 & 3 & -6 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & -8 & 3 & 4 \\ -9 & 2 & 3 & 4 \\ 1 & 2 & -7 & 4 \\ 1 & 2 & 3 & -6 \end{pmatrix} \\ &\rightarrow \begin{pmatrix} 1 & -8 & 3 & 4 \\ 0 & -7 & 3 & 4 \\ 0 & 1 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & -8 & 3 & 4 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \end{aligned}$$

得 $\alpha_1, \alpha_2, \alpha_3$ 为极大线性无关组, 且 $\alpha_4 = -\alpha_1 - \alpha_2 - \alpha_3$.

方法点评: 本题考查向量组线性相关性性质及非齐次线性方程组的解.

当向量的个数与维数相等时, 该向量组线性相关的充分必要条件是該向量组构成的行列式为零.

(21)【解】 (I) 由 $A \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$, 得 $\lambda_3 = 3$ 为 A 的特征值, $\alpha_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ 为其对应的特征向量.

因为 $AX=0$ 有非零解, 所以 $\lambda=0$ 为 A 的特征值, 其对应的特征向量为

$$\alpha_1 = \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}, \alpha_2 = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix},$$

因为 α_1, α_2 线性无关, 所以 $\lambda=0$ 为 A 的二重特征值,

于是 $\lambda_1 = \lambda_2 = 0$, 其对应的线性无关的特征向量为 α_1, α_2 .

$$(II) \text{ 令 } \beta_1 = \alpha_1 = \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}, \beta_2 = \alpha_2 - \frac{(\alpha_2, \beta_1)}{(\beta_1, \beta_1)} \beta_1 = \frac{1}{2} \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \beta_3 = \alpha_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix},$$

$$\text{单位化得} \quad \gamma_1 = \frac{1}{\sqrt{6}} \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}, \gamma_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \gamma_3 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix},$$

$$\text{令 } Q = \begin{pmatrix} -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \\ \frac{2}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \end{pmatrix}, \text{ 则 } Q^T A Q = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

$$(III) \quad A = Q \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 3 \end{pmatrix} Q^T = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix},$$

$$\begin{aligned} \left(A - \frac{3}{2}E\right)^6 &= \left[Q\left(A - \frac{3}{2}E\right)Q^T\right]^6 = Q\left(A - \frac{3}{2}E\right)^6 Q^T \\ &= Q \begin{pmatrix} \left(\frac{3}{2}\right)^6 & 0 & 0 \\ 0 & \left(\frac{3}{2}\right)^6 & 0 \\ 0 & 0 & \left(\frac{3}{2}\right)^6 \end{pmatrix} Q^T = \left(\frac{3}{2}\right)^6 E. \end{aligned}$$

(22)【解】 (I) $F_Y(y) = P\{Y \leq y\} = P\{X^2 \leq y\}$,

当 $y < 0$ 时, $F_Y(y) = 0$;

当 $0 \leq y < 1$ 时, $F_Y(y) = P\{-\sqrt{y} \leq X \leq \sqrt{y}\}$

$$= \int_{-\sqrt{y}}^0 \frac{1}{2} dx + \int_0^{\sqrt{y}} \frac{1}{4} dx = \frac{3\sqrt{y}}{4};$$

当 $1 \leq y < 4$ 时, $F_Y(y) = P\{-\sqrt{y} \leq X \leq \sqrt{y}\} = P\{-1 \leq X \leq \sqrt{y}\}$

$$= \int_{-1}^0 \frac{1}{2} dx + \int_0^{\sqrt{y}} \frac{1}{4} dx = \frac{1}{2} + \frac{\sqrt{y}}{4};$$

当 $y \geq 4$ 时, $F_Y(y) = 1$,

$$\text{于是 } F_Y(y) = \begin{cases} 0, & y < 0, \\ \frac{3\sqrt{y}}{4}, & 0 \leq y < 1, \\ \frac{1}{2} + \frac{\sqrt{y}}{4}, & 1 \leq y < 4, \\ 1, & y \geq 4, \end{cases} \quad \text{故 } f_Y(y) = \begin{cases} \frac{3}{8\sqrt{y}}, & 0 < y < 1, \\ \frac{1}{8\sqrt{y}}, & 1 < y < 4, \\ 0, & \text{其他.} \end{cases}$$

$$(II) E(X) = \int_{-1}^0 \frac{x}{2} dx + \int_0^2 \frac{x}{4} dx = \frac{1}{4},$$

$$E(Y) = E(X^2) = \int_{-1}^0 \frac{x^2}{2} dx + \int_0^2 \frac{x^2}{4} dx = \frac{5}{6},$$

$$E(XY) = E(X^3) = \int_{-1}^0 \frac{x^3}{2} dx + \int_0^2 \frac{x^3}{4} dx = \frac{7}{8},$$

$$\text{则 } \text{Cov}(X, Y) = E(XY) - E(X) \cdot E(Y)$$

$$= E(X^3) - E(X) \cdot E(X^2) = \frac{7}{8} - \frac{5}{24} = \frac{2}{3}.$$

$$(\text{III}) F\left(-\frac{1}{2}, 4\right) = P\left\{X \leqslant -\frac{1}{2}, X^2 \leqslant 4\right\} = P\left\{-1 \leqslant X \leqslant -\frac{1}{2}\right\} = \int_{-1}^{-\frac{1}{2}} \frac{1}{2} dx = \frac{1}{4}.$$

$$\begin{aligned} (23) \text{【解】} \quad (\text{I}) E(X) &= \int_{-\infty}^{+\infty} x f(x) dx = \int_0^1 \theta x dx + \int_1^2 (1-\theta)x dx \\ &= \frac{\theta}{2} + \frac{3(1-\theta)}{2} = \frac{3}{2} - \theta, \end{aligned}$$

$$\text{令 } E(X) = \bar{X}, \text{ 得参数 } \theta \text{ 的矩估计量为 } \hat{\theta} = \frac{3}{2} - \bar{X}.$$

(II) 似然函数为

$$\begin{aligned} L(\theta) &= P\{X_1 = x_1\} P\{X_2 = x_2\} \cdots P\{X_n = x_n\} \\ &= P\{X = x_1\} P\{X = x_2\} \cdots P\{X = x_n\} \\ &= \theta^N (1-\theta)^{n-N}, \end{aligned}$$

$$\ln L(\theta) = N \ln \theta + (n-N) \ln(1-\theta),$$

$$\text{令 } \frac{d}{d\theta} \ln L(\theta) = \frac{N}{\theta} - \frac{n-N}{1-\theta} = 0 \text{ 得参数 } \theta \text{ 的最大似然估计为 } \hat{\theta} = \frac{N}{n}.$$